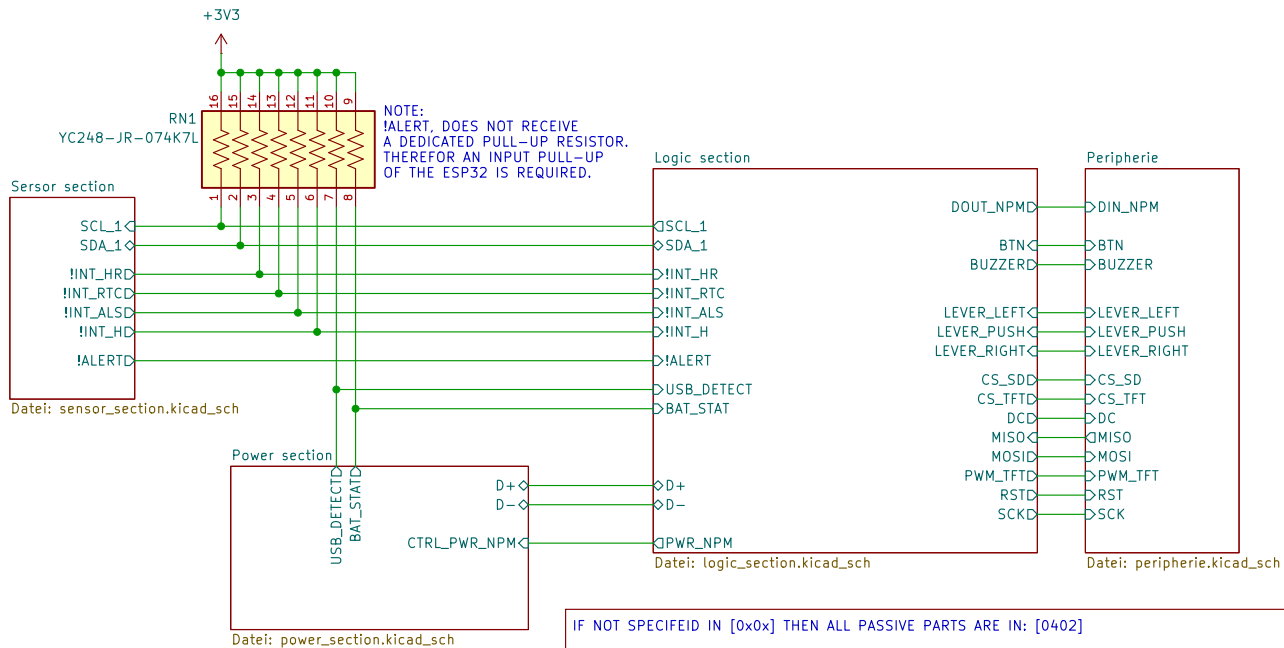


Pull-Ups



IF NOT SPECIFEID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /
File: NX-ISD.kicad_sch

Title:

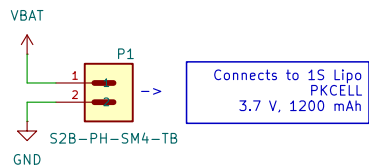
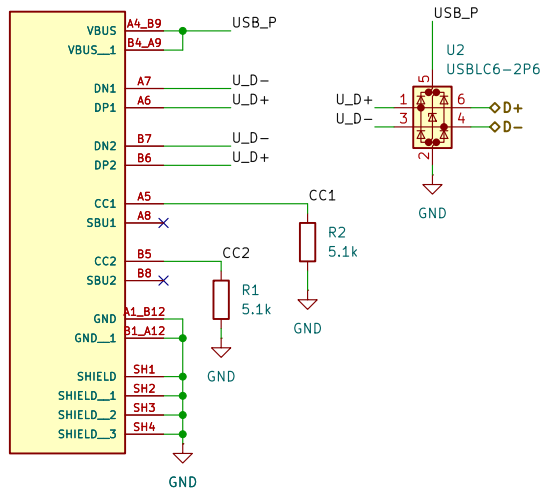
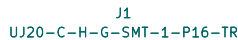
Size: A5
KiCad E.D.A. 10.0.5

Date:

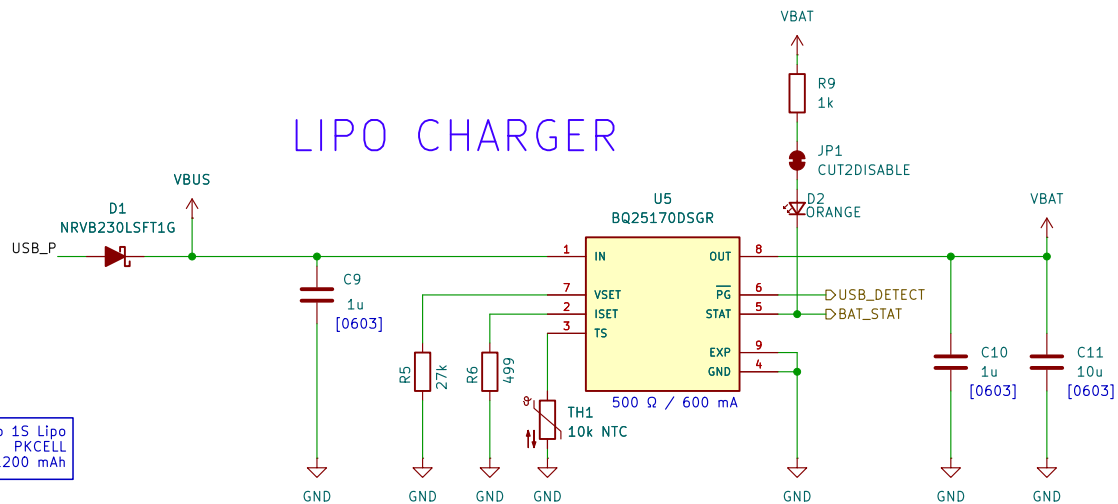
Rev:

Id: 1/5

CONNECTORS



LIPO CHARGER

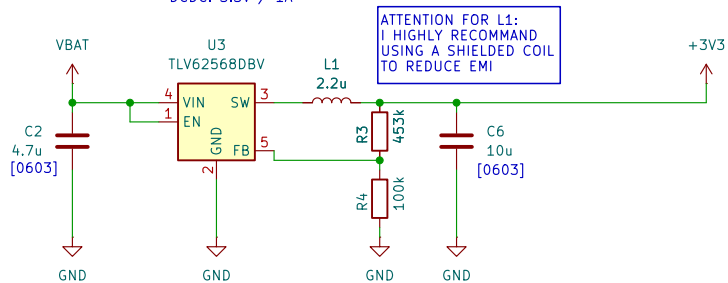


Formula R_ISET: $300 / I_{OUT}(\text{in Amperes})$

Charge Current	Calculated R_{ISET}
50 mA	6000 Ω
100 mA	3000 Ω
150 mA	2000 Ω
200 mA	1500 Ω
300 mA	1000 Ω
400 mA	750 Ω
500 mA	600 Ω
600 mA	500 Ω
800 mA	375 Ω

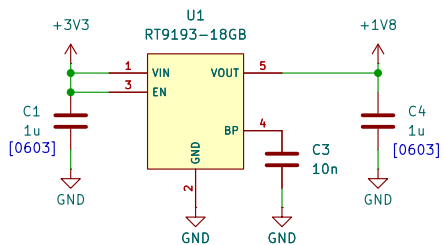
+3V3 DCDC Buck

DCDC: 3.3V / 1A



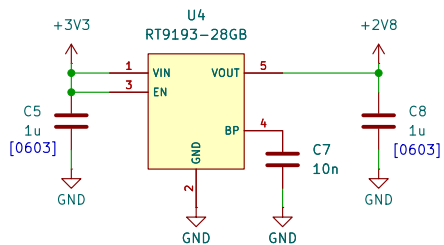
+1V8 LDO

(USED FOR HRM)

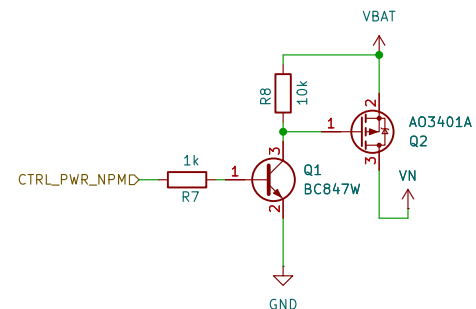


+2V8 LDO

(USED FOR DISPLAY)



NPM POWER CONTROL



IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Power section/
File: power_section.kicad_sch

Title:

Size: A4	Date:
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KiCad E.D.A. 10.0.5

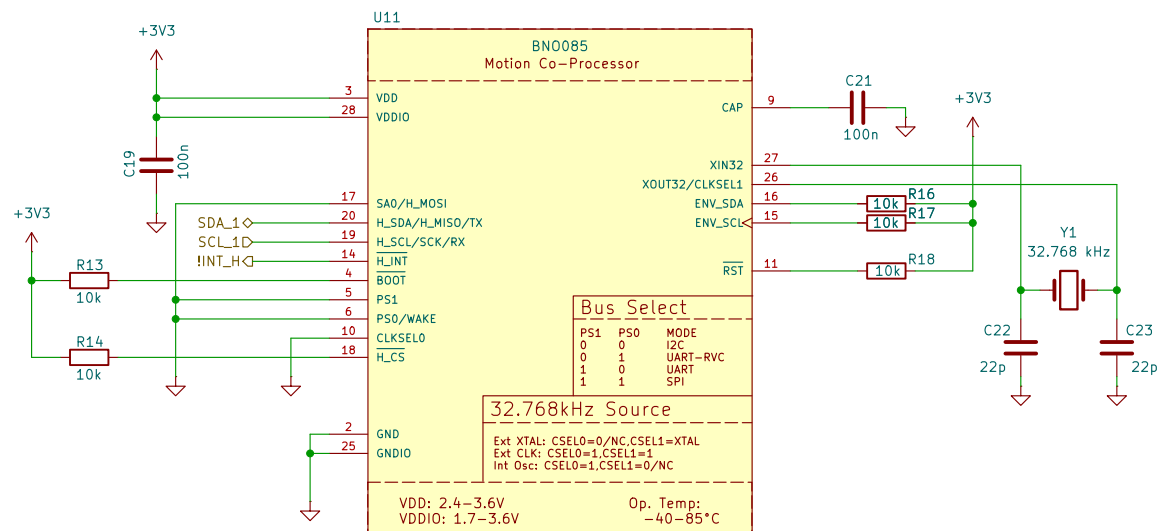
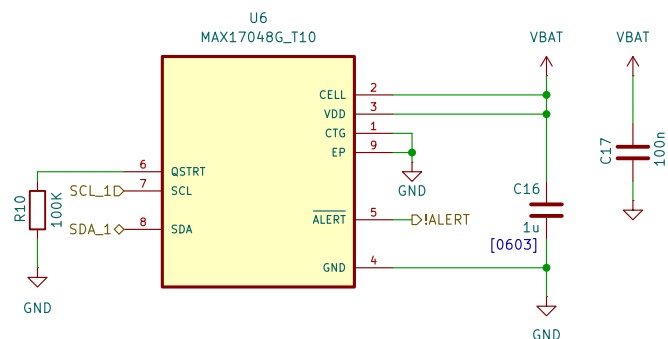
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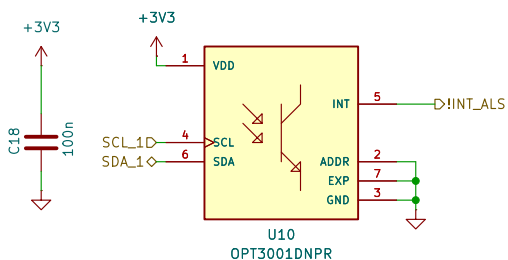
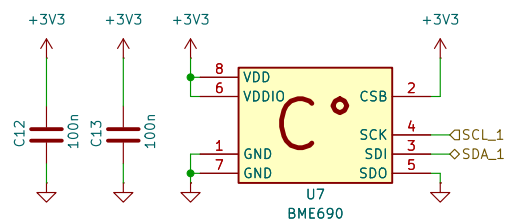
Rev:

Id: 2/5

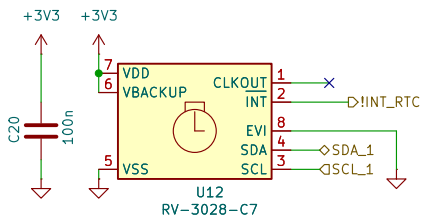
9-DOF



Ambient Light Sensor



RTC



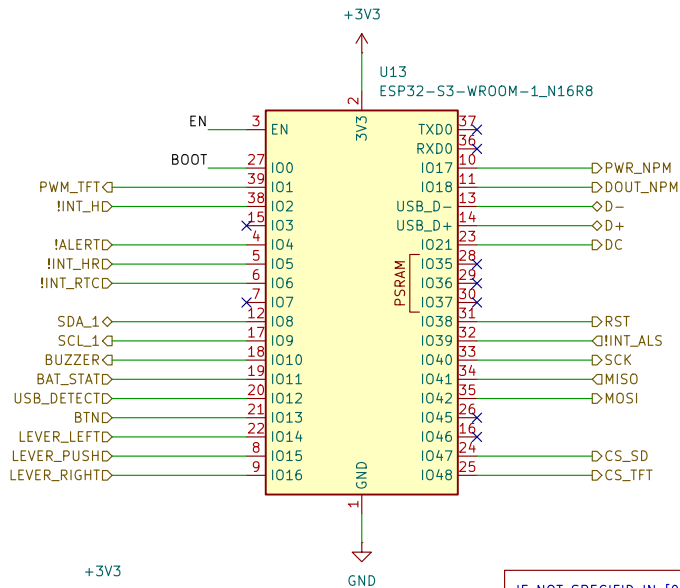
The schematic shows the connection between the MAX30102 (U8) and PCA9306 (U9). The MAX30102 is powered by +3V3 and +1V8 rails. Its VDD (pin 11), VLED+ (pin 10), and VLED- (pin 9) are connected to +3V3, +1V8, and GND (pin 12) respectively. A 1µF capacitor (C14) is connected across +3V3 and GND, and a 100nF capacitor (C15) is connected across +1V8 and GND. The MAX30102's INT pin (pin 13) is connected to the INT_HR signal. The I2C pins SDA (pin 3) and SCL (pin 2) are connected to the PCA9306's SDA1 (pin 4) and SCL1 (pin 3) respectively. The PCA9306 is powered by +1V8 and its EN pin (pin 7) is connected to +1V8 through a 200k resistor (R15). The PCA9306's VREF1 (pin 2) and VREF2 (pin 7) pins are connected to each other and to GND (pin 1). The PCA9306's output pins SDA_1 (pin 8) and SCL_1 (pin 6) are shown as open connections.

Sheet: /Sersor section/
File: sensor_section.kicad_sch

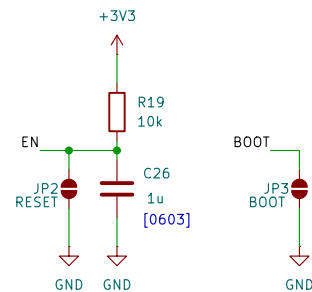
Size: A4	Date:
KiCad E.D.A. 10.0.5	

Rev:
Id: 3/5

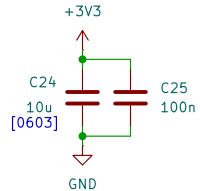
ESP32-S3 N16R8



EN & BOOT



ESP32-S3 chip strapping pins:
GPIO 0.
GPIO 3.
GPIO 45.
GPIO 46.



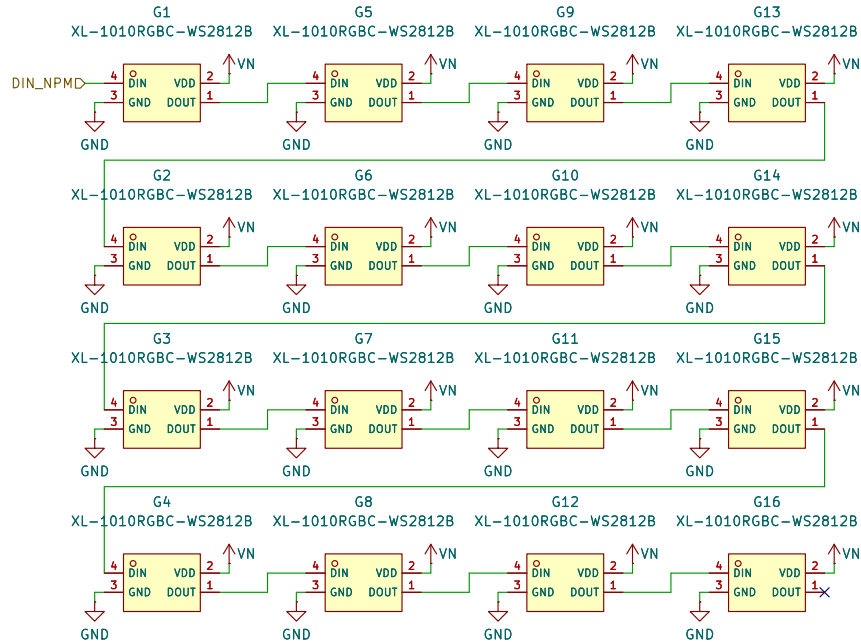
IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Logic section/
File: logic_section.kicad_sch

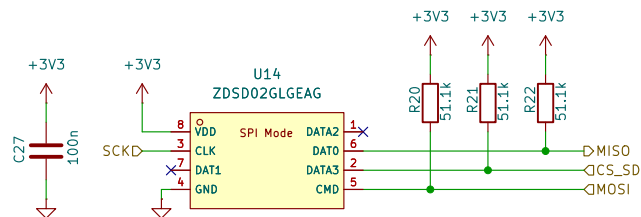
Title:		Rev:	
Size: A5	Date:	Id: 4/5	
KiCad E.D.A. 10.0.5			

NeoPixel Matrix (NPM)

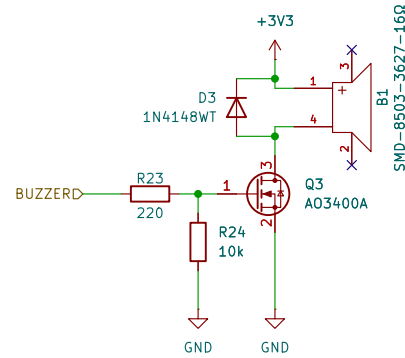
(NPM DOES NOT REQUIRE DECOUPLING CAPACITORS DUE TO THE SHORT PATH OF ITS SUPPLY)



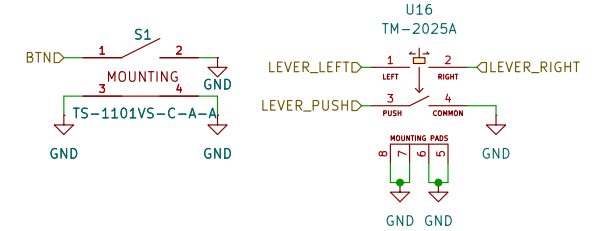
NAND-SD



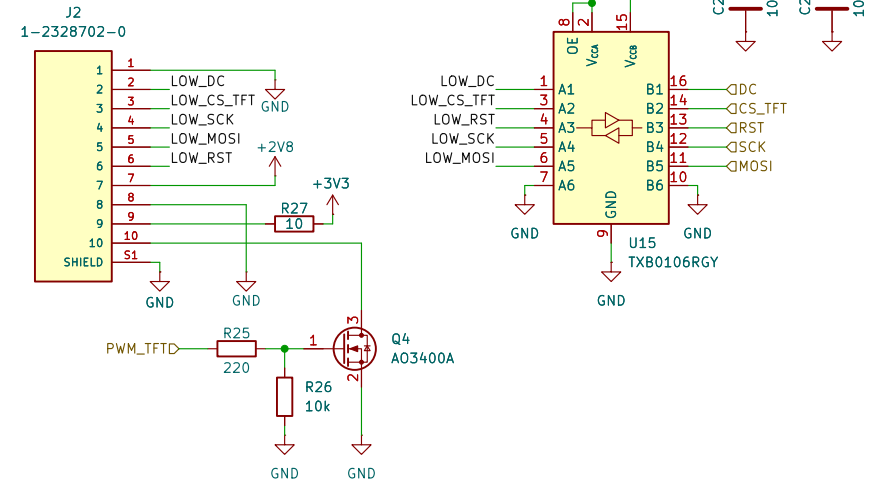
BUZZER



BUTTONS



DISPLAY



Sheet: /Peripherie/
File: peripherie.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 10.0.5

Rev:

Id: 5/5